

FRANCIS SESAY

AI and ML Solutions Developer

PROFESSIONAL SUMMARY

AI and Machine Learning Solutions Developer | Educator | Founder, Sierra Edutech Solutions | Technical Lead, KNESST SL Ltd

As a first-class IT graduate and the best graduating student at IPAM, University of Sierra Leone, I have accumulated over 5 years of experience in driving AI-powered innovations, full-stack software development, and education technology initiatives throughout Sierra Leone. I am proficient in Python (Flask, Django), PHP, Next.js, Node.js, and Tailwind CSS, and I have advanced experience in database design (MySQL, PostgreSQL) and the deployment of **scalable, production-grade web applications**. Skilled in leveraging **AI-assisted development tools** to build **end-to-end intelligent systems** that enhance learning, employability, and business performance. As **founder of Sierra Edutech Solutions** and **technical lead at KNESST SL Ltd.**, I have mentored over 3,000 learners while leading programmes that raise **technology awareness, digital inclusion, and innovation capacity** nationwide. Actively engaged in **national development initiatives** with partners such as DSTI and MOCTI, I am passionate about creating **machine learning platforms, data systems, and digital learning ecosystems** that expand access, empower communities, and drive sustainable progress across Sierra Leone.

EMPLOYMENT HISTORY

TEACHING ASSISTANT

Feb 2025 – Present

Institute of Public Administration and Management (IPAM), University of Sierra Leone Freetown, Sierra Leone

- ◆ Conduct weekly lab sessions and guest lectures for **100+ undergraduate IT students**, focusing on applied concepts in AI, data analytics, and visualization.
- ◆ Design and implement **hands-on exercises** using real-world datasets, such as **Freetown traffic data** and **local health statistics**, to strengthen students' analytical and problem-solving skills.
- ◆ **Mentor final-year capstone teams**, providing guidance on project design, data analysis methodologies, and presentation delivery — resulting in **over 90% of students achieving distinction-level grades**.

TECHNICAL LEAD

Jul 2024 – Present

KNESST SL Ltd

Freetown, Sierra Leone

- ◆ Leading initiatives to empower tech builders and entrepreneurs, focusing on foundational tech skills and scalable product development.
- ◆ Spearheaded the Capacity Building Initiative, enhancing career readiness for school leavers.
- ◆ Facilitated training programmes that upskilled participants in domains like Agri-tech, Women in Tech, Fintech, AI and education.
- ◆ Organized the Developers Challenge Initiative, fostering scalable tech solutions for local challenges.
- ◆ Built and deployed a job-ready platform named Harya for connecting job seekers to employers.
- ◆ Facilitated Developers challenge national hackathons, guiding teams from ideation to execution of solutions and leading them into incubation period to connect them to investors to enhance national

development.

- ◆ Worked with leading tech government institutions in Sierra Leone such as DSTI, MOCTI, Christex Foundation for national development project.

TECHNICAL LEAD

Jul 2023 – Nov 2024

Google Developer Student Club (GDSC), University of Sierra Leone

- ◆ Delivered workshops on coding, machine learning, and web development, reaching over 50 students.
- ◆ Supported participants in creating innovative projects for the GDSC Solution Challenge.
- ◆ Improved student proficiency in web development and Python through tailored educational materials.

PRESIDENT, FACULTY OF INFORMATION SYSTEMS AND TECHNOLOGY

Sep 2022 – Sep 2023

IPAM, University of Sierra Leone

- ◆ Launched a mentorship program supporting over 800 students, leading to a 90% improvement in grades.
- ◆ Organized 15+ technical workshops on programming, AI, and software development.
- ◆ Established the Ishmail-Fomba Legacy Awards, recognizing 50 students for excellence in academics and leadership.
- ◆ Introduced peer tutoring programs, boosting participation and academic outcomes for over 1000+ students

MASTER PEER TUTOR

Mar 2022 – Apr 2024

IPAM Writing Centre

- ◆ I provided training and mentorship to students in academic writing, which resulted in enhanced performance throughout their academic journey at university of Sierra Leone.
- ◆ Reviewed over 300 essays, provided detailed feedback, and organized workshops for students on campus as a master tutor.
- ◆ Supervised and coached upcoming peer tutors to expand the program's reach.

TECHNICAL AND CLIENT RELATIONS SPECIALIST

Dec 2022 – Present

Kugba Host

- ◆ Secured 10 new clients, increasing the company's client base by 15%.
- ◆ Managed website development projects, improving online presence for non-profits.
- ◆ Provided technical support, optimizing client satisfaction and web functionality.

DIRECTOR AND TUTOR

Nov 2021 – Dec 2024

Leotech SL

- ◆ Led the Skills Against Poverty program, training over 150 students in web development and Python.
- ◆ Developed a scalable educational model addressing youth unemployment through tech education.

FOUNDER AND TECHNICAL LEAD

Dec 2019 – Present

Sierra Edutech Solution SL Ltd

- ◆ Mentored over 3,000 students, with 90% achieving improved academic results.
- ◆ Launched skills training programs in Python, web development, and AI, enhancing job market readiness.
- ◆ Managed a team of 10 tutors and expanded operations across multiple institutions.

- ◆ Built and deployed company website with content management system using HTML, CSS, JavaScript, TailwindCSS, and PHP.
- ◆ Building Brainbox, an e-learning platform for university students in Sierra Leone to access past questions and AI resources.

EDUCATION

B. SC. HONORS IN INFORMATION TECHNOLOGY (FIRST CLASS) Nov 2019 – Apr 2023

Institute of Public Administration and Management (IPAM), University of Sierra Leone *Sierra Leone*

CGPA: 3.83/4.0 (Top 1% of a cohort of 2,600 graduates).

Awards: Best Graduating Student, Best Dissertation (AI-Based Inquiry Chatbot – Score: 80%).

Relevant Coursework: Intelligent Systems, Biometrics, Database Systems, System Analysis, Object-Oriented Programming, Human-Computer Interaction, Information Security, Computer Networking

PROFESSIONAL TRAINING & CERTIFICATIONS

- ◆ DevOps and Software Engineering | *Coursera (Ongoing)*
- ◆ IBM Data Science & Analyst Professional Certificate | *Coursera (Ongoing)*
- ◆ Data Analysis with Python | *IBM Cognitive Class*
- ◆ Responsive Website Development | *University of London & Coursera*
- ◆ Certified Peer Tutor | *IPAM, University of Sierra Leone*
- ◆ Google Cyber Security | *Coursera (Ongoing)*
- ◆ Mern stack front to back: full stack react, redux & node Js | *Coursera*
- ◆ Data engineering, big data, and machine learning on GCP specialization
Coursera

RESEARCH AND PROJECTS

AI-Based Inquiry Chatbot

- ◆ Developed an NLP-powered chatbot using GPT-3.5 Turbo, LangChain, and Chroma DB to resolve over 80% of administrative queries, providing real-time support to students and reducing workload for staff.
- ◆ Designed an ingestion pipeline adaptable for multiple domains, including education, healthcare, and agriculture, based on the data it is trained on.
- ◆ Awarded Best Final Year Dissertation (2024).

Automated CV Screening Tool for HR Departments

- ◆ Designed a CV screening solution using GPT-3.5 Turbo, LangChain, and Chroma DB, streamlining candidate matching to job requirements in resource-constrained environments.

Traffic Congestion Prediction System

- ◆ Built an end-to-end machine learning project using a dummy dataset with historical traffic data from the streets of Freetown.
- ◆ Developed a predictive model to forecast traffic in specific areas based on environmental and temporal factors.
- ◆ Designed the system to leverage historical trends for actionable insights, supporting improved traffic management in urban areas.

Facial Recognition Attendance System

- ◆ Designed and deployed an AI-driven facial recognition system for attendance tracking in corporate and educational settings.
- ◆ Implemented using concepts from the FaceNet paper, "FaceNet: A Unified Embedding for Face Recognition and Clustering," which optimizes face embeddings for efficient recognition.
- ◆ Developed an end-to-end pipeline leveraging triplet loss for improved face similarity measures, ensuring high accuracy and scalability.

Crop Yield Prediction System

- ◆ Developed a comprehensive crop yield prediction system tailored to the needs of smallholder farmers.

- ◆ Built a crop recommendation application to predict the most suitable crops based on soil and environmental data.
- ◆ Created a fertilizer recommendation application suggesting optimal fertilizer combinations to enhance soil nutrient management.
- ◆ Designed a plant disease prediction application to identify plant diseases from leaf images and provide treatment recommendations.

SKILLS

Python (*Experienced*), Java (*Experienced*), HTML (*Experienced*), CSS (*Experienced*), JavaScript (*Experienced*), PHP (*Experienced*), Flask (*Experienced*), Django (*Experienced*), Full-Stack Development (*Experienced*), REST APIs (*Experienced*), Responsive Design (*Experienced*), Machine Learning (*Experienced*), Data Analysis (*Experienced*), Pandas (*Experienced*), NumPy (*Experienced*), Matplotlib (*Experienced*), AWS (*Skillful*), Docker (*Skillful*), CI/CD Pipelines (*Skillful*), MySQL (*Experienced*), SQLite (*Experienced*), Postgres (*Experienced*), Agile, Scrum, Microservices Architecture.

LINKS

LinkedIn: [linkedin.com](https://www.linkedin.com), Website: sierraedutech.com.

ADDITIONAL INFORMATION

AWARDS & HONORS

- ◆ Overall Best Graduating Student (Class of 2023): Faculty of Information Systems, IPAM.
- ◆ Best Dissertation Project (2024): AI-Based Inquiry Chatbot.
- ◆ Leadership Excellence Award (2024): Recognized for transformative initiatives in the Faculty of Information Systems.
- ◆ 30 A' s Minimum Distinction Award (2023): For achieving distinctions in 30+ modules.
- ◆ Best Programmer (2023): Faculty of Information Systems, IPAM.
- ◆ Africell Impact Foundation Award (2024): Overall Best Graduating Student, Faculty of Information Systems.

COMMUNITY ENGAGEMENT

- ◆ KNESST Community: Trained 15+ women in digital skills, promoting gender inclusion in tech.
- ◆ Ruth Charity Foundation: Developed a website attracting donors, supporting education for underserved children.
- ◆ Progressive Syndicate Tutorial Classes: Tutored over 200 students, achieving an 80% pass rate.

PRESENTATIONS & SEMINARS

- ◆ World Quality Week (2023): Delivered a presentation on quality management in ICT, engaging over 100 participants.
- ◆ Write and Win Symposium (2023): Panelist offering strategies for academic and professional writing to 200+ students.

REFERENCES

Abass Conteh, UNIVERSITY OF SIERRA LEONE (aconteh@ipam.edu.sl, +23277341870);

Mohamed Kugba Massaquoi, KUGBA HOST SL LTD (mohamed.massaquoi@kugbahost.com, +23279006790).